

weeks-1-and-2\2nd-math-problems.py

```
1 '''
2 simplifying the square root of a whole number < 100
3 '''
4  $\sqrt{32}$ 
5
6 #find the largest perfect square factor ---> 16
7  $32 = 16 * 2$ 
8
9 #rewrite square root of 32 as square root of (16 * 2)
10
11  $\sqrt{32} = \sqrt{(16 * 2)}$ 
12
13 #separate the square roots
14  $\sqrt{32} = \sqrt{16} * \sqrt{2}$ 
15
16 #simplify  $\sqrt{16}$ 
17  $\sqrt{32} = 4 * \sqrt{2}$ 
18  $4 * \sqrt{2}$ 
19
20
21 '''
22 applying the negative exponent rule to a fraction
23 '''
24  $(3 / 8)^{-2}$ 
25 '''
26
27 negative exponent rule
28  $(a / b)^{-n} = (b / a)^n = (b^2 / a^2)$ 
29 '''
30
31 #swap the numerator and denominator and change the negative exponent to be positive
32  $(3 / 8)^{-2} =$ 
33  $(8 / 3)^2$ 
34
35 #distribute the exponent to both the numerator and denominator
36  $(8 / 3)^2 =$ 
37  $(8^2 / 3^2) =$ 
38  $(64 / 9)$ 
39
40
41 '''
42 applying the negative exponent rule to a fraction
43 '''
44  $(4 / 3)^{-1}$ 
45 '''
46
47 negative exponent rule
48  $(a / b)^{-n} = (b / a)^n = (b^2 / a^2)$ 
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49 '''
50
51 #swap the numerator and denominator and change the negative exponent to be positive
52  $(4 / 3)^{-1} =$ 
53  $(3 / 4)^1$ 
54
55 #distribute the exponent to both the numerator and denominator
56  $(3 / 4)^2 =$ 
57
58
59 '''
60 applying the negative exponent rule to a fraction
61 '''
62  $(4 / 5)^{-3}$ 
63 '''
64
65 negative exponent rule
66  $(a / b)^{-n} = (b / a)^n = (b^2 / a^2)$ 
67 '''
68
69 #swap the numerator and denominator and change the negative exponent to be positive
70  $(5 / 4)^3 =$ 
71
72 #distribute the exponent to both the numerator and denominator
73  $((5)^3 / (4)^3) =$ 
74  $(125 / 64)$ 
75
76 #it doesn't simplify further than this because 125 and 64 have no common factors
77
78
79 '''
80 applying the negative exponent rule to a fraction
81 '''
82  $(7 / 2)^{-2}$ 
83
84 '''
85 negative exponent rule
86  $(a / b)^{-n} = (b / a)^n = (b^2 / a^2)$ 
87 '''
88
89 #swap the numerator and denominator and change the negative exponent to be positive
90  $(2 / 7)^2$ 
91
92 #distribute the exponent to both the numerator and denominator
93  $((2)^2 / (7)^2) =$ 
94  $(4 / 49)$ 
95
96 #it doesn't simplify further than this because 4 and 49 have no common factors
97
98 '''
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99  _____
100 applying the power of 0 rule to various expressions
101 '''
102 - (6)0
103 (- (1 / 7))0
104
105 '''
106 power of 0 rule: anything nonzero raised to the power of 0 becomes 1
107 (a)0 = 1
108 '''
109
110 - (6)0 =
111 - (1) =
112 -1
113
114 (- (1 / 7))0 =
115 1
116
117 #everything inside the parentheses (including the negative sign) is raised to the power of 0,
118 so the whole thing becomes 1
119
120 _____
121 applying the power of 0 rule to various expressions
122 '''
123 - (5 / 7)0
124 80
125
126 '''
127 power of 0 rule: anything nonzero raised to the power of 0 becomes 1
128 (a)0 = 1
129 '''
130
131 - (5 / 7)0 =
132 - (1) =
133 -1
134
135 80 =
136 1
137
138 #everything inside the parentheses (including the negative sign) is raised to the power of 0,
139 so the whole thing becomes 1
140
141 _____
142 applying the power of 0 rule to various expressions
143 '''
144 40
145 (-3) (2 / 5)0
146

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147 '''
148 power of 0 rule: anything nonzero raised to the power of 0 becomes 1
149 (a)0 = 1
150 '''
151
152 40 =
153 1
154
155 (-3) (2 / 5)0 =
156 (-3) (1) =
157 -3
158
159 # everything inside the parentheses (including the negative sign) is raised to the power of
160 0, so the whole thing becomes 1
161
162 _____
163 simplifying expressions with negative exponents
164 '''
165 (-2)-3
166
167 '''
168 negative exponent rule
169 (a / b)-n = (b / a)n = (b2 / a2)
170 '''
171
172 #swap the numerator and denominator and change the negative exponent to be positive
173 #keep the integer base negative since we only got rid of the negative exponent from using the
174 'negative exponent rule'
175
176 (-2)-3 =
177 1 / (-2)3 =
178
179 1 / (-8) =
180 -(1 / 8)
181
182 _____
183 simplifying expressions with negative exponents
184 '''
185 (-5)-2
186
187 '''
188 negative exponent rule
189 (a / b)-n = (b / a)n = (b2 / a2)
190 '''
191
192 #swap the numerator and denominator and change the negative exponent to be positive
193 #keep the integer base negative since we only got rid of the negative exponent from using the
194 'negative exponent rule'
195
196 (-5)-2 =
197 1 / (-5)2 =

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195
196 1 / 25
197
198 #it doesn't simplify further than this because 1 and 25 have no common factors
199
200 '''
201
202 simplifying expressions with negative exponents
203 '''
204  $(-3)^{-2}$ 
205
206 '''
207 negative exponent rule
208  $(a / b)^{-n} = (b / a)^n = (b^2 / a^2)$ 
209 '''
210
211 #swap the numerator and denominator and change the negative exponent to be positive
212 #keep the integer base negative since we only got rid of the negative exponent from using the
    'negative exponent rule'
213  $(-3)^{-3} =$ 
214  $1 / (-3)^3 =$ 
215
216
217  $1 / (-27) =$ 
218  $-(1 / 27)$ 
219 #it doesn't simplify further than this because 1 and 27 have no common factors
220
221 '''
222
223 simplifying expressions with negative exponents
224 '''
225  $(-8)^{-2}$ 
226
227 '''
228 negative exponent rule
229  $(a / b)^{-n} = (b / a)^n = (b^2 / a^2)$ 
230 '''
231
232 #swap the numerator and denominator and change the negative exponent to be positive
233 #keep the integer base negative since we only got rid of the negative exponent from using the
    'negative exponent rule'
234  $(-8)^{-2} =$ 
235  $1 / (-8)^2 =$ 
236  $1 / 64$ 
237
238 #it doesn't simplify further than this because 1 and 64 have no common factors
239
240 '''
241
242 simplifying trinomials with negative coefficients, and variables raised to a power

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243 '''
244  $((-4) * (y)^2 * (x))^3 =$ 
245
246 #distribute the exponent to each factor inside the parentheses → →
247  $(-4)^3 * ((y)^2)^3 * (x)^3 =$ 
248
249 #multiply doubled-up exponents together, so  $((y)^2)^3$  becomes  $(y)^6$  → →
250  $(-64) * (y)^6 * (x)^3$ 
251
252 #this can't be simplified further because there are no like terms or common factors
253
254 '''
255
256 simplifying trinomials with negative coefficients, and variables raised to a power
257 '''
258  $((-7) * (y) * (x)^3)^2 =$ 
259
260 #distribute the exponent to each factor inside the parentheses → →
261  $(-7)^2 * (y)^2 * ((x)^3)^2 =$ 
262
263 #multiply doubled-up exponents together → →
264  $(49) * (y)^2 * (x)^6 =$ 
265  $49 y^2 x^6$ 
266
267 #this can't be simplified further because there are no like terms or common factors
268
269 '''
270
271 simplifying trinomials with negative coefficients, and variables raised to a power
272 '''
273  $(2 * (w)^2 * (v)^3)^5 =$ 
274
275 #distribute the exponent to each factor inside the parentheses → →
276  $(2)^5 * ((w)^2)^5 * ((v)^3)^5 =$ 
277
278 #multiply doubled-up exponents together → →
279  $(32) * ((w)^{10}) * ((v)^{15}) =$ 
280  $32 w^{10} v^{15}$ 
281
282 #this can't be simplified further because there are no like terms or common factors
283
284 '''
285
286 simplifying trinomials with negative coefficients, and variables raised to a power
287 '''
288  $((-5) * (u)^2 * (w))^3$ 
289
290 #distribute the exponent to each factor inside the parentheses → →
291  $(-5)^3 * ((u)^2)^3 * (w)^3 =$ 
292

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293 #multiply doubled-up exponents together → →
294 (-125) * (u)6 * (w)3 =
295 -125 u6 w3
296
297 #this can't be simplified further because there are no like terms or common factors
298
299 '''
300 _____
301 finding the degree and leading coefficient of a polynomial
302 '''
303 (6)(x)3 + (20)(x)8 + (-2) + (20)(x) =
304
305 #arrange the terms so the variables with exponents go from left to right in descending order
306 (20)(x)8 + (6)(x)3 + (20)(x) + (-2) =
307
308 #the leading coefficient is before the first term in the polynomial when arranged in standard
form
309 (20)
310
311 #the degree is the power/exponent of the leading coefficient
312 (8)
313
314 '''
315 _____
316 finding the degree and leading coefficient of a polynomial
317 '''
318 (-4)(w)7 + (20)(w)9 + (-w) + (-20)(w)8
319
320 #arrange the terms so the variables with exponents go from left to right in descending order
321 (20)(w)9 + (-20)(w)8 + (-4)(w)7 + (-w)
322
323 #the leading coefficient is before the first term in the polynomial when arranged in standard
form
324 (20)
325
326 #the degree is the power/exponent of the leading coefficient
327 (9)
328
329
330 '''
331 _____
332 finding the degree and leading coefficient of a polynomial
333 '''
334 18 + 2u + (-1)(u)8
335
336 #arrange the terms so the variables with exponents go from left to right in descending
order(20)(w)9 + (-20)(w)8 + (-4)(w)7 + (-w)
337 (-1)(u)8 + 2u + 18
338
339 #the leading coefficient is the first term in the polynomial when arranged in standard form
340 (-1)

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341
342 #the degree is the power/exponent of the leading coefficient
343 8
344
345
346 '''
347 _____
348 finding the degree and leading coefficient of a polynomial
349 '''
350 (-20)(w)7 + 15 + (-2)(w) + (-15)(w)8
351
352 #arrange the terms so the variables with exponents go from left to right in descending order
353 (-15)(w)8 + (-20)(w)7 + (-2)(w) + 15
354
355 #the leading coefficient is before the first term in the polynomial when arranged in standard
form
356 (-15)
357
358 #the degree is the power/exponent of the leading coefficient
359 (8)
360
361
362 '''
363 simplifying the square root of a perfect square monomial
364 '''
365 √((49)(w)4)
366
367 #separate the square roots of each factor inside the radical
368 #perfect squares: 4, 9, 16, 25, 36, 49, ...
369 √(49) * √((w)4) =
370 7 * (w)2
371
372
373 '''
374 simplifying the square root of a perfect square monomial
375 '''
376 √((81)(z)10)
377
378 #separate the square roots of each factor inside the radical
379 #perfect squares: 4, 9, 16, 25, 36, 49, ...
380 √(81) * √((z)10) =
381 9 * (z)10/2
382
383 #when you take the square root of a power, you divide the power by 2
384 9 * (z)5 =
385 9z5
386
387
388 '''
389 simplifying the square root of a perfect square monomial

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390 '''
391  $\sqrt{(16)(w)^8}$ 
392
393 #separate the square roots of each factor inside the radical
394 #perfect squares: 4, 9, 16, 25, 36, 49, ...
395  $\sqrt{16} * \sqrt{(w)^8} =$ 
396  $4 * (w)^{8/2}$ 
397
398 #when you take the square root of a power, you divide the power by 2
399  $4 * (w)^4 =$ 
400  $4w^4$ 
401
402
403 '''
404 applying 'power of a power' rule and 'negative exponent' rule
405 '''
406  $((x)^5)^{-7}$ 
407
408 #power of a power' rule: multiply the exponents
409  $(x)^5 * (-7) =$ 
410  $(x)^{-35} =$ 
411
412 #negative exponent' rule:  $a^{-n} = (1 / a^n)$ 
413  $(1 / (x)^{35})$ 
414
415
416 '''
417 applying 'power of a power' rule and 'negative exponent' rule
418 '''
419  $((u)^{-2})^7$ 
420
421 #power of a power' rule: multiply the exponents
422  $(u)^{-2} * (7) =$ 
423  $(u)^{-14} =$ 
424
425 #negative exponent' rule:  $a^{-n} = (1 / a^n)$ 
426  $(1 / (u)^{14})$ 
427
428
429 '''
430 applying 'rules of exponents' rule and 'power of a power' rule using fraction exponents
431 '''
432  $(16)^{5/4}$ 
433
434  $(16)^{5/4} =$ 
435  $((16)^{1/4})^5 =$ 
436
437 #an exponent of  $(1/4)$  means the exact same thing as taking the fourth root of a number
438  $(\sqrt[4]{16})^5 =$ 
439

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440  $\#2 * 2 * 2 * 2 = 16$ 
441  $(2)^5 =$ 
442 32
443
444
445 '''
446 next problem
447 '''
448  $(16)^{(3/2)} =$ 
449  $((16)^{1/2})^3 =$ 
450
451 #an exponent of  $(1/2)$  means the exact same thing as taking the square root of a number
452  $(\sqrt{16})^3 =$ 
453  $(4)^3 =$ 
454 (64)
455
456 '''
457 next problem
458 '''
459  $(64)^{(2/3)} =$ 
460
461 #use the 'power of a power' rule **backwards**:
462 #for the power  $(2/3)$  we can write it as  $(1/3) * (2)$ :
463  $((64)^{1/3}) * (2) =$ 
464  $((64)^{1/3})^2$ 
465
466 #an exponent of  $(1/3)$  means the exact same thing as taking the cube root of a number:
467  $(\sqrt[3]{64})^2 =$ 
468  $(4)^2 =$ 
469 (16)
470
471 '''
472 power of a power rule, power of a product rule, and power of a quotient rule
473 '''
474  $((-8)(a) / (b)^3)^2 =$ 
475  $((-8)(a))^2 / (b^3)^2 =$ 
476
477 #square both the coefficient and the variable in the numerator:
478  $((-8)^2 * (a)^2) / (b^3)^2 =$ 
479  $((64)(a)^2) / (b^3)^2 =$ 
480
481 #multiply the exponents:
482  $(64)(a)^2 / (b)^6 =$ 
483
484 #submission version with no extra formatting:
485  $64a^2/b^6$ 
486
487 '''
488 power of a power rule, power of a product rule, and power of a quotient rule
489 '''

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```

490 ((v)4 / (-4)(w)3)3 =
491
492 #raise all terms inside the gold parenthesis to the power outside of it
493 ((v)4)3 / ((-4)3 * ((w)3)3) =
494
495 #multiply the exponents:
496 ((v)12) / ((-64)(w)9) =
497
498 #move the negative sign to the numerator:
499 -((v)12) / ((64)(w)9) =
500
501 #submission version with no extra formatting:
502 -v12/64w9
503
504 '''
505 power of a power rule, power of a product rule, and power of a quotient rule
506 '''
507 ((2)(x)4 / (y))5 =
508
509 #raise all terms inside the gold parenthesis to the power outside of it
510 ((2)5 * (x)4)5 / (y)5 =
511
512 #multiply the exponents and evaluate the coefficient:
513 ((32)(x)20) / (y)5 =
514
515 #submission version with no extra formatting:
516 32x20/y5
517
518 '''
519 radical to exponent rule
520 '''
521 √17 =
522
523 #a square root means raising the number to the power of (1/2)
524 (17)1/2 =
525
526 #submission version with no extra formatting:
527 171/2
528
529 '''
530 exponent to radical rule
531 '''
532 (t)3/4 =
533
534 #the denominator (4) is the root index, and the numerator (3) is the exponent
535 (4√(t))3 or 4√(t3)
536
537 #either form is mathematically correct because the order of multiplication between the root
538 and exponent does not change the result

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539 '''
540 exponent to radical rule
541 '''
542 (13)2/5 =
543
544 #the denominator (5) is the root index, and the numerator (2) is the exponent
545 (5√(13))2 or 5√(132)
546
547 #either form is mathematically correct because the order of multiplication between the root
548 and exponent does not change the result

```